Applied Math II: Statistical Learning

Midterm exam, 2026.02.10

Name:

You have 80 minutes to complete the exam. You may use a sheet of notes, but no computers or textbooks are allowed. Please write legibly, and indicate clearly if your solution is continued on another sheet of paper. Good luck!

1. Independence vs. zero covariance

Let X, Y be real-valued random variables with finite second moments.

1. (3 points) Prove: if X and Y are independent, then $\mathrm{cov}(X, Y) = 0$.
2. (3 points) Give an explicit example where $\mathrm{cov}(X, Y) = 0$ but X and Y are **not** independent.

2. Bias–variance decomposition for a scalar estimator

Let $\hat{\theta}$ be an estimator of a scalar parameter θ based on data D. For example, the data could be $D = \{x_1, \ldots, x_N\}$, where $x_n \in \mathbb{R}$, and $\hat{\theta}$ could be the ML or MAP estimate for the mean of this dataset. Define the mean of the parameter estimates across datasets generated from the same distribution as $\mu_{\hat{\theta}} = E_D[\hat{\theta}]$.

1. (3 points) Prove the identity

$$E_D[(\hat{\theta} - \theta)^2] = (\mu_{\hat{\theta}} - \theta)^2 + \mathrm{Var}_D(\hat{\theta}). \tag{1}$$

2. (3 points) Interpret the two terms in one sentence each, in the context of “overfitting vs.

underfitting."

3. Linear transform of a multivariate Gaussian

A student recalls: if $x \sim \mathcal{N}(\mu, \Sigma)$ and $z = Ax + b$, then z is Gaussian. They code:

```
def affine_gaussian(mu, Sigma, A, b):
    mu_z = A @ mu + b
    Sigma_z = A @ Sigma @ A
    return mu_z, Sigma_z
```

(3 points) Identify the bug and correct it.

4. Bayesian linear regression posterior

(5 points) Assume the model $t_n = \mathbf{w}^\top \boldsymbol{\phi}_n + \epsilon_n$ with $\epsilon_n \sim \mathcal{N}(0, \beta^{-1})$, and assume that the data points are statistically independent from one another. Let Φ have rows $\boldsymbol{\phi}_n^\top$ and let the prior be $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1} I)$.

Derive the posterior $p(\mathbf{w} \mid \mathbf{t}) = \mathcal{N}(\mathbf{m}_N, S_N)$, and identify S_N^{-1} and $\mathbf{m}_N$.

5. Generative Gaussian classifier $\Rightarrow$ logistic sigmoid

(6 points) Two classes with Gaussian class-conditionals sharing covariance Σ:

$$p(\mathbf{x} \mid C_k) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \Sigma), \quad p(C_1) = \pi, \; p(C_2) = 1 - \pi. \tag{2}$$

Show that

$$p(C_1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + w_0) \tag{3}$$

where $\sigma(a) = 1/(1 + e^{-a})$ is the logistic sigmoid function, and give $\mathbf{w}$ and w_0.

6. Divergence in MLE for logistic regression

(3 points) In 2–4 sentences, explain why, if the data are perfectly linearly separable, the MLE for logistic regression typically satisfies $\|\mathbf{w}\| \to \infty$.